Lesson 17: Radians, Arcs & Sectors — Practice Worksheet

Name: ______________________________ Date: ____________

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Convert 90° to radians (exact).

2. Convert 60° to radians (exact).

3. Convert π/4 to degrees.

4. Convert 180° to radians.

5. Convert 3π/2 to degrees.

6. Convert 45° to radians (exact).

Section 2 — Core practice

7. Arc length: $r = 8$ cm, $\theta = 0.5$ rad.

8. Sector area: $r = 6$ cm, $\theta = 1.5$ rad.

9. Arc length: $r = 12$ cm, $\theta = 0.25$ rad.

10. Sector area: $r = 10$ cm, $\theta = 0.8$ rad.

11. Convert 150° to radians (exact).

12. Arc length: $r = 5$ cm, $\theta = 2$ rad.

Section 3 — Challenge & exam-style

13. A sector has radius 9 cm and angle 1.2 rad. Find the arc length and area.

14. A sector has area 24 cm^2 and radius 6 cm. Find the angle θ (radians).

15. A sector has arc length 15 cm and radius 10 cm. Find θ.

16. Find the perimeter of a sector with $r = 7$ cm and $\theta = 1$ rad.